

Kilonovae and *R*-process Nucleosynthesis from Neutron Star Mergers

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FEBRUARY 26, 2026

Outline

1. Introduction
 - Question: what is the astrophysical origin of *heavy* elements?
 - Answer: Neutron star mergers?
2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)
3. Modelling Astrophysics: How much *r*-process is produced? (mass)
4. Conclusions

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The origin of elements

																		13		14		15		16		17		18	
																		IIIA		IVA		VA		VIA		VIIA		VIII	
																		5		6		7		8		9		10	
																		B		C		N		O		F		Ne	
																		Boron 10.81 2.3		Carbon 12.011 2.4		Nitrogen 14.007 2.5		Oxygen 15.999 2.6		Fluorine 18.998 2.7		Neon 20.180 2.8	
																		3		4		5		6		7		8	
																		Li		Be									
																		Lithium 6.94 2.1		Beryllium 9.012 2.2									
																		11		12		13		14		15		16	
																		Na		Mg		Al		Si		P		S	
																		Sodium 22.98976928 2.81		Magnesium 24.305 2.82		Aluminum 26.982 2.83		Silicon 28.085 2.84		Phosphorus 30.974 2.85		Sulfur 32.06 2.86	
																		19		20		21		22		23		24	
																		K		Ca		Sc		Ti		V		Cr	
																		Potassium 39.0983 2.863		Calcium 40.078 2.862		Scandium 44.955908 2.862		Titanium 47.867 2.862		Vanadium 50.9415 2.862		Chromium 51.9961 2.862	
																		37		38		39		40		41		42	
																		Rb		Sr		Y		Zr		Nb		Mo	
																		Rubidium 85.468 2.863-8.1		Strontium 87.62 2.863-2		Yttrium 88.90584 2.863-2		Zirconium 91.224 2.863-2		Niobium 92.90637 2.863-2		Molybdenum 95.95 2.863-2	
																		55		56		57-71		72		73		74	
																		Cs		Ba				Hf		Ta		W	
																		Cesium 132.90545196 2.863-8.1		Barium 137.327 2.863-8.1				Hafnium 178.49 2.863-32-2		Tantalum 180.94788 2.863-32		Tungsten 183.84 2.863-32-2	
																		87		88		89-103		104		105		106	
																		Fr		Ra				Rf		Db		Sg	
																		Francium 223 2.863-37-8.1		Radium 226 2.863-32-8.1				Rutherfordium 261 2.863-32-32-2		Dubnium 262 2.863-32-32-12		Seaborgium 266 2.863-32-32-32-2	
																		61		62		63		64		65		66	
																		Pm		Sm		Eu		Gd		Tb		Dy	
																		Promethium 145 2.863-25-2		Samarium 150.36 2.863-2		Europium 151.96 2.863-2		Gadolinium 157.25 2.863-2		Terbium 158.93 2.863-2		Dysprosium 162.50 2.863-2	
																		93		94		95		96		97		98	
																		Np		Pu		Am		Cm		Bk		Cf	
																		Neptunium 237 2.863-32-32-2		Plutonium 244 2.863-32-32-8.2		Americium 243 2.863-32-32-2		Curium 247 2.863-32-32-2		Berkelium 247 2.863-32-32-2		Californium 251 2.863-32-32-2	
																		107		108		109		110		111		112	
																		Bh		Hs		Mt		Ds		Rg		Cn	
																		Bohrium 264 2.863-32-32-32-2		Hassium 277 2.863-32-32-32-2		Meitnerium 268 2.863-32-32-32-2		Darmstadtium 271 2.863-32-32-32-1		Roentgenium 272 2.863-32-32-32-2		Copernicium 285 2.863-32-32-32-2	
																		113		114		115		116		117		118	
																		Nh		Fl		Mc		Lv		Ts		Og	
																		Nihonium 284 2.863-32-32-32-2		Flerovium 289 2.863-32-32-32-4		Moscovium 288 2.863-32-32-32-2		Livermorium 293 2.863-32-32-32-4		Tennessine 294 2.863-32-32-32-8		Oganesson 294 2.863-32-32-32-8	
																		57		58		59		60		61		62	
																		La		Ce		Pr		Nd		Pm		Sm	
																		Lanthanum 138.91 2.863-6		Cerium 140.12 2.863-6		Praseodymium 140.91 2.863-6		Neodymium 144.24 2.863-6		Promethium 145 2.863-6		Samarium 150.36 2.863-6	
																		89		90		91		92		93		94	
																		Ac		Th		Pa		U		Np		Pu	
																		Actinium 227 2.863-32-32-8.2		Thorium 232.04 2.863-32-32-8.2		Protactinium 231.04 2.863-32-32-8.2		Uranium 238.03 2.863-32-32-8.2		Neptunium 237 2.863-32-32-8.2		Plutonium 244 2.863-32-32-8.2	
																		95		96		97		98		99		100	
																		Am		Cm		Bk		Cf		Es		Fm	
																		Americium 243 2.863-32-32-8.2		Curium 247 2.863-32-32-8.2		Berkelium 247 2.863-32-32-8.2		Californium 251 2.863-32-32-8.2		Einsteinium 252 2.863-32-32-8.2		Fermium 257 2.863-32-32-8.2	
																		101		102		103		104		105		106	
																		Md		No		Lr							
																		Mendelevium 258 2.863-32-32-8.2		Nobelium 259 2.863-32-32-8.2		Lawrencium 262 2.863-32-32-8.2							

State of matter (color of name)
GAS LIQUID SOLID UNKNOWN

Subcategory in the metal-metalloid-nonmetal trend (color of background)
Alkali metals Lanthanides Metalloids
Alkaline earth metals Actinides Reactive nonmetals
Transition metals Post-transition metals Noble gases

Unknown chemical properties

The origin of elements

 H ¹ Hydrogen	 The big bang																 He ² Helium															
 Li ³ Lithium	 Be ⁴ Beryllium	 Dying low-mass stars																 White dwarf supernovae	 Radioactive decay													
 Na ¹¹ Sodium	 Mg ¹² Magnesium	 Cosmic ray collisions																 Dying high-mass stars	 Merging neutron stars	 Human-made												
 K ¹⁹ Potassium	 Ca ²⁰ Calcium	 Sc ²¹ Scandium	 Ti ²² Titanium	 V ²³ Vanadium	 Cr ²⁴ Chromium	 Mn ²⁵ Manganese	 Fe ²⁶ Iron	 Co ²⁷ Cobalt	 Ni ²⁸ Nickel	 Cu ²⁹ Copper	 Zn ³⁰ Zinc	 Ga ³¹ Gallium	 Ge ³² Germanium	 As ³³ Arsenic	 Se ³⁴ Selenium	 Br ³⁵ Bromine	 Kr ³⁶ Krypton															
 Rb ³⁷ Rubidium	 Sr ³⁸ Strontium	 Y ³⁹ Yttrium	 Zr ⁴⁰ Zirconium	 Nb ⁴¹ Niobium	 Mo ⁴² Molybdenum	 Tc ⁴³ Technetium	 Ru ⁴⁴ Ruthenium	 Rh ⁴⁵ Rhodium	 Pd ⁴⁶ Palladium	 Ag ⁴⁷ Silver	 Cd ⁴⁸ Cadmium	 In ⁴⁹ Indium	 Sn ⁵⁰ Tin	 Sb ⁵¹ Antimony	 Te ⁵² Tellurium	 I ⁵³ Iodine	 Xe ⁵⁴ Xenon															
 Cs ⁵⁵ Cesium	 Ba ⁵⁶ Barium																	 Hf ⁷² Hafnium	 Ta ⁷³ Tantalum	 W ⁷⁴ Tungsten	 Re ⁷⁵ Rhenium	 Os ⁷⁶ Osmium	 Ir ⁷⁷ Iridium	 Pt ⁷⁸ Platinum	 Au ⁷⁹ Gold	 Hg ⁸⁰ Mercury	 Tl ⁸¹ Thallium	 Pb ⁸² Lead	 Bi ⁸³ Bismuth	 Po ⁸⁴ Polonium	 At ⁸⁵ Astatine	 Rn ⁸⁶ Radon
 Fr ⁸⁷ Francium	 Ra ⁸⁸ Radium																	 Rf ¹⁰⁴ Rutherfordium	 Db ¹⁰⁵ Dubnium	 Sg ¹⁰⁶ Seaborgium	 Bh ¹⁰⁷ Bohrium	 Hs ¹⁰⁸ Hassium	 Mt ¹⁰⁹ Meitnerium	 Ds ¹¹⁰ Darmstadtium	 Rg ¹¹¹ Roentgenium	 Cn ¹¹² Copernicium	 Nh ¹¹³ Nihonium	 Fl ¹¹⁴ Flerovium	 Mc ¹¹⁵ Moscovium	 Lv ¹¹⁶ Livermorium	 Ts ¹¹⁷ Tennessine	 Og ¹¹⁸ Oganesson
		 La ⁵⁷ Lanthanum		 Ce ⁵⁸ Cerium	 Pr ⁵⁹ Praseodymium	 Nd ⁶⁰ Neodymium	 Pm ⁶¹ Promethium	 Sm ⁶² Samarium	 Eu ⁶³ Europium	 Gd ⁶⁴ Gadolinium	 Tb ⁶⁵ Terbium	 Dy ⁶⁶ Dysprosium	 Ho ⁶⁷ Holmium	 Er ⁶⁸ Erbium	 Tm ⁶⁹ Thulium	 Yb ⁷⁰ Ytterbium	 Lu ⁷¹ Lutetium															
		 Ac ⁸⁹ Actinium		 Th ⁹⁰ Thorium	 Pa ⁹¹ Protactinium	 U ⁹² Uranium	 Np ⁹³ Neptunium	 Pu ⁹⁴ Plutonium	 Am ⁹⁵ Americium	 Cm ⁹⁶ Curium	 Bk ⁹⁷ Berkelium	 Cf ⁹⁸ Californium	 Es ⁹⁹ Einsteinium	 Fm ¹⁰⁰ Fermium	 Md ¹⁰¹ Mendelevium	 No ¹⁰² Nobelium	 Lr ¹⁰³ Lawrencium															

NASA's Goddard Space Flight Center

Density of objects in the Universe

Water



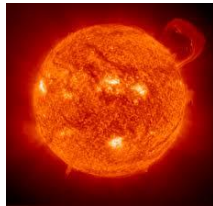
$$1 \text{ g cm}^{-3}$$

Earth



$$2 - 15 \text{ g cm}^{-3}$$

Sun



$$10 - 200 \text{ g cm}^{-3}$$

Neutron Star



$$? \text{ g cm}^{-3}$$

Density of objects in the Universe

A. $1,000 \text{ g cm}^{-3}$

B. $100,000 \text{ g cm}^{-3}$

C. $10,000,000 \text{ g cm}^{-3}$

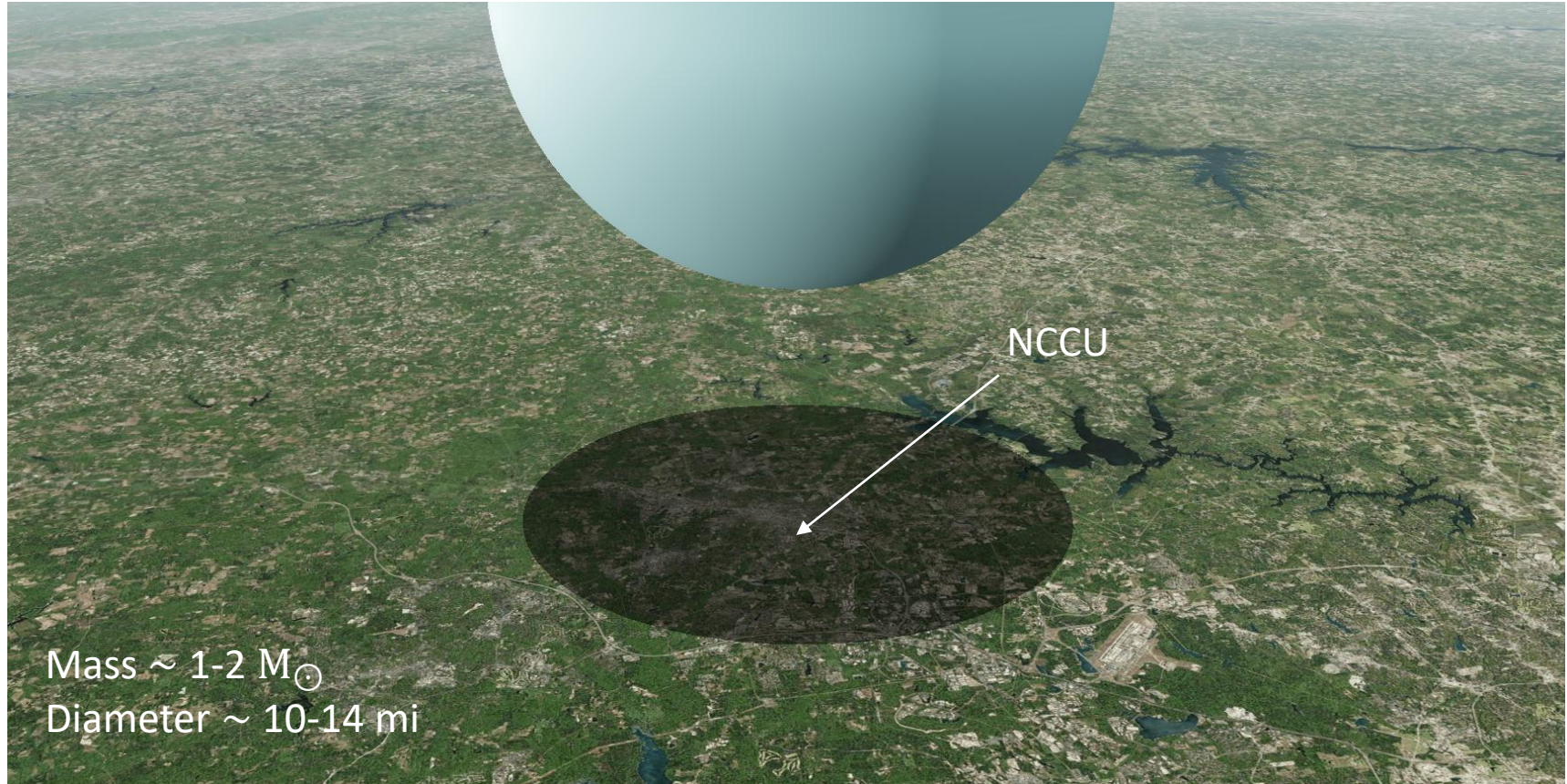
D. $1,000,000,000,000,000 \text{ g cm}^{-3} \sim 10^{15}$ ✓

Neutron Star



$10^{15} \text{ g cm}^{-3}$

Neutron star over Durham



Neutron star merger



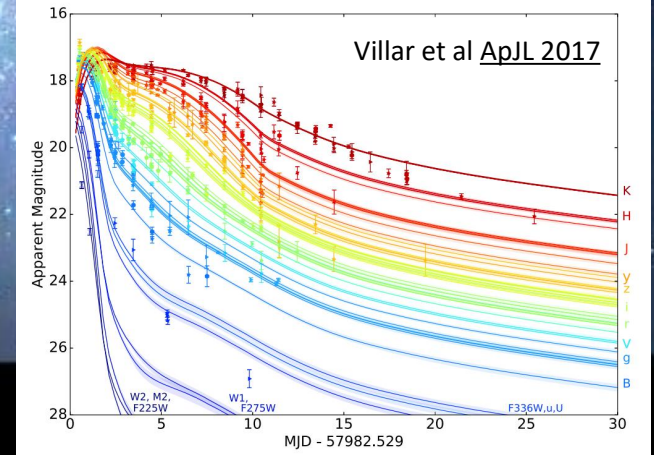
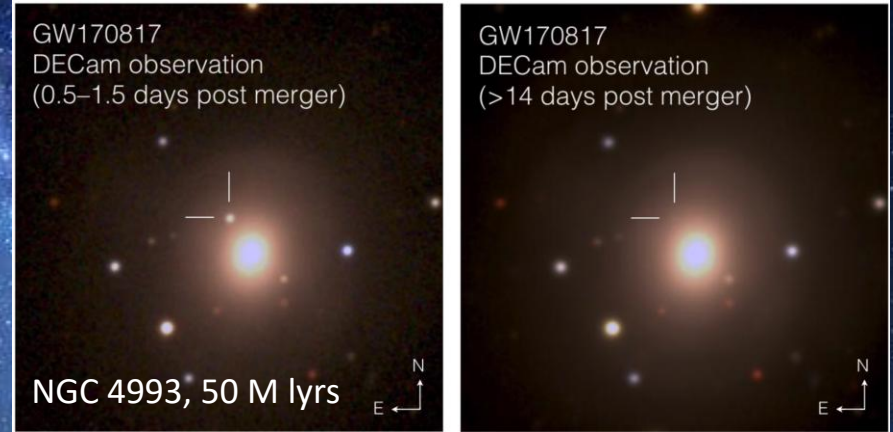
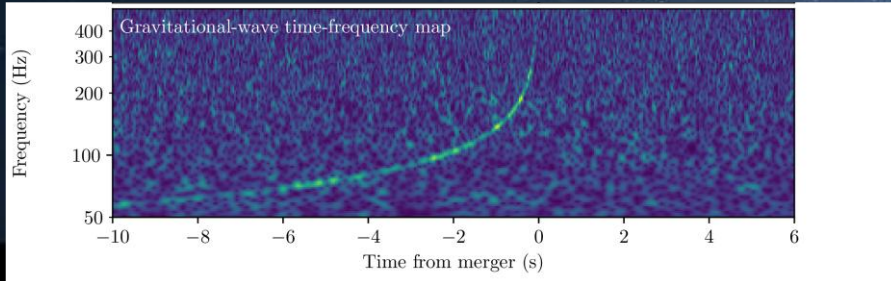
Artistic illustration from [NASA Goddard](#)

Neutron star merger: GW170817/AT2017gfo



Two perpendicular 2.5 mile arms making
a Michaelson-Morley setup

LIGO-Livingston-Louisiana, LIGO-Hanford-Washington,
Virgo-Italy, KARGA-Japan, upcoming LIGO-India (2030s)



Modelling a neutron star merger

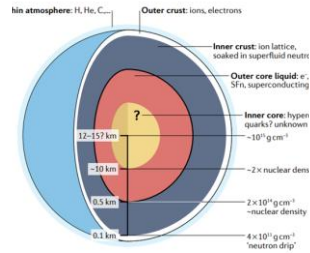
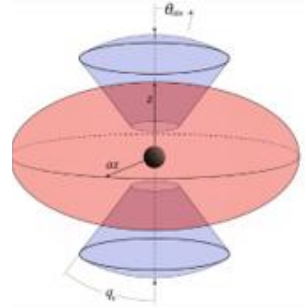


Fig. 1 | Schematic of the structure of a neutron star and its internal structure. The diagram illustrates the thin atmosphere, the outer and inner crust, and the outer and inner core, with the respective densities at different depths. Adapted with permission from the NICER Team.

Neutron star structure

Equation of state

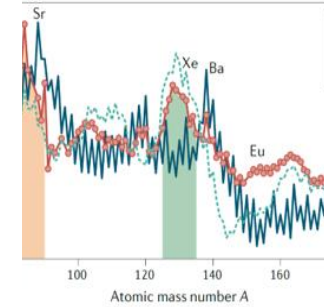
Bovard+17, Radice+18, Prakash+22,
Kedia+22, Kedia+24, Hensh+24



Hydrodynamics

General Relativistic

Breschi+21, Bulla+19,23, Heinzl+21,
Kawaguchi+20,21, **Kedia+23, Fryer+24**



Nuclear reactions

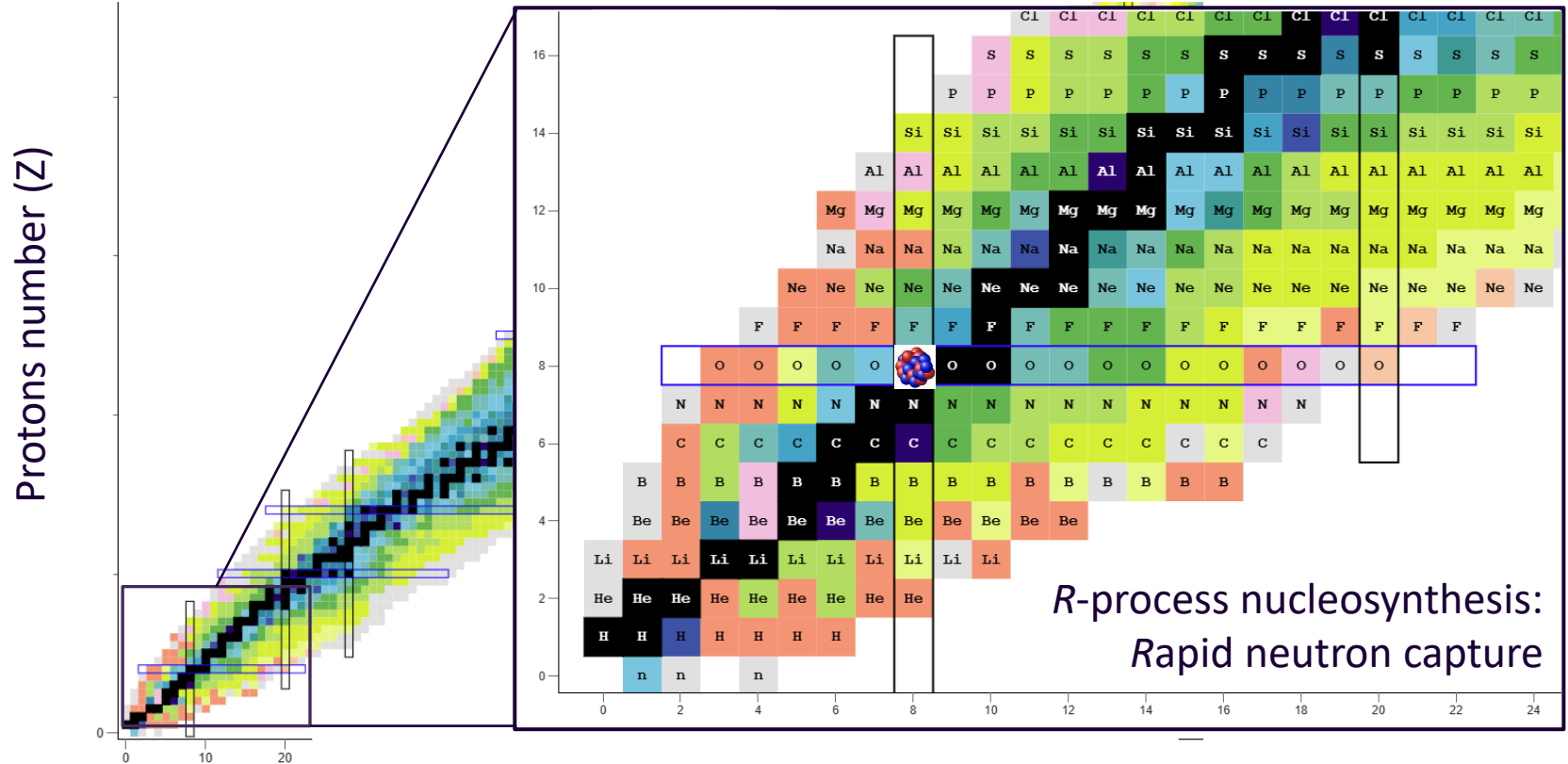
r-process nucleosynthesis

Mumpower+16, Vassh+18,+20,
Zhu+18,Holmbeck+23, **Kedia+26**

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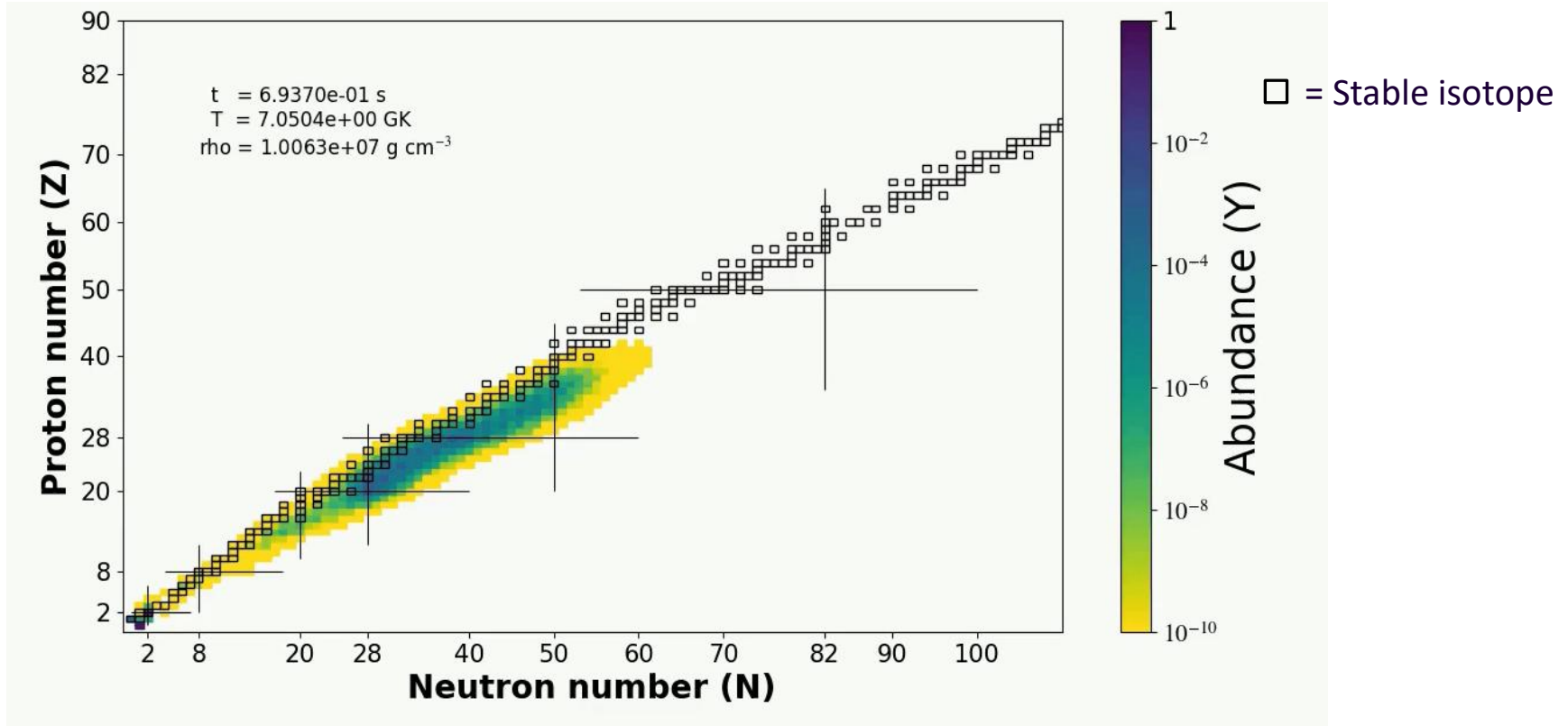
R-process on the Chart of Nuclides



Neutron number (N)

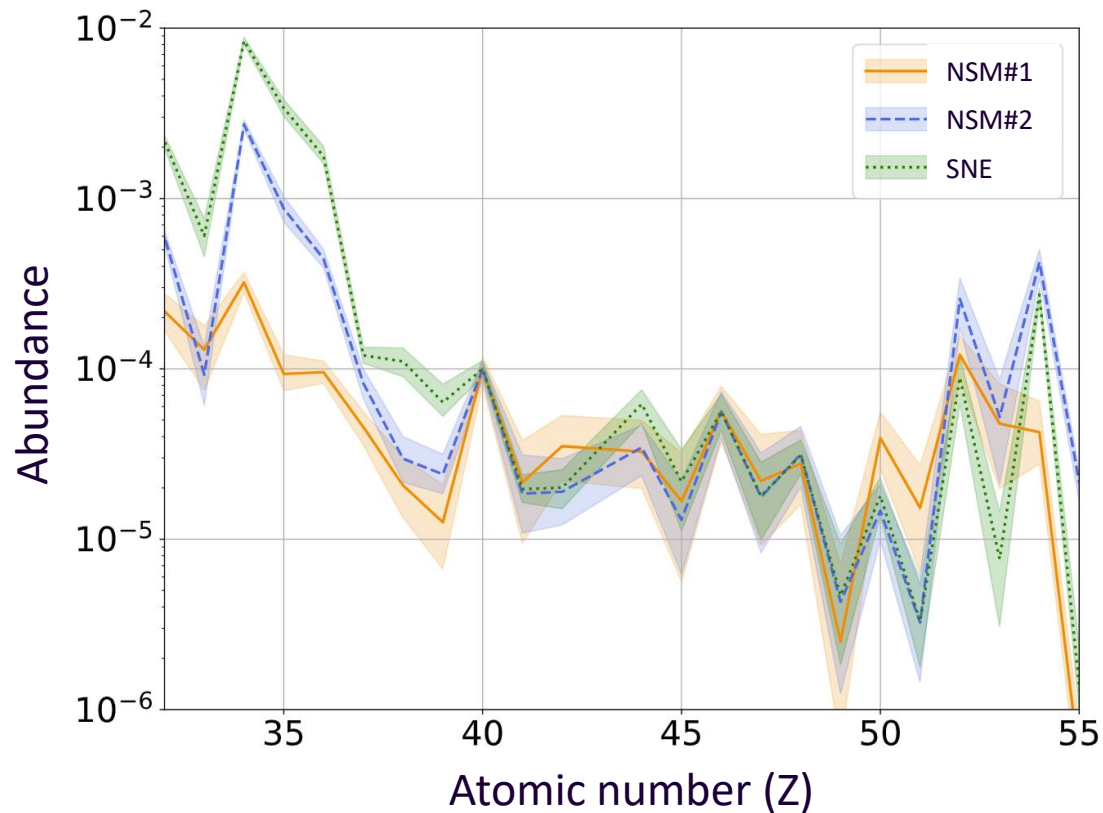
Representation of all Stable and unstable isotopes

R-process evolution



Movie by AK; Calculation done on **PRISM** (reaction network code)

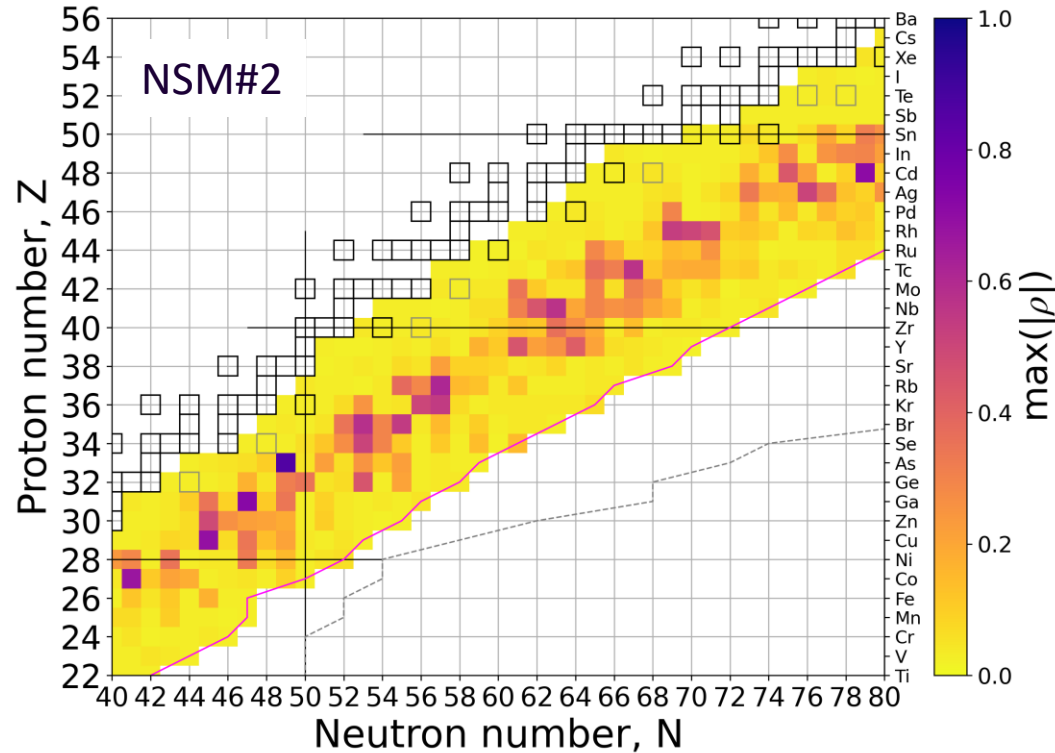
Uncertainties in r -process abundances



2σ (95 percentile) uncertainty in abundances due to uncertainties in theoretical neutron capture reaction rates.

*Shown is only the weak r -process abundance pattern.

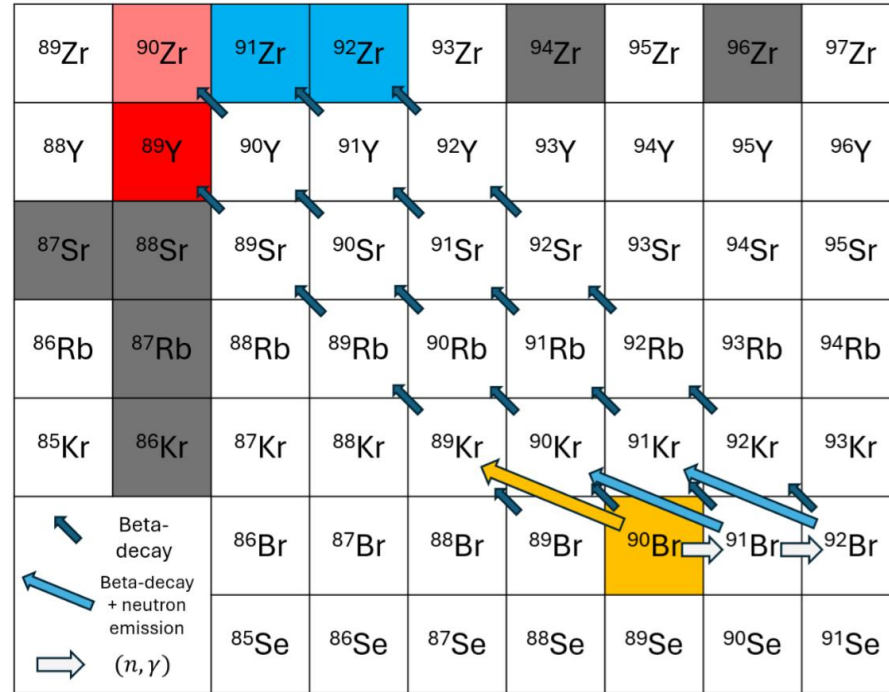
Most influential isotopes for r process



Element	(n, γ) isotope, correlation coeff. (MF14)			
Se	^{78}Ga -0.65	^{80}Ga -0.33	-	-
Br	^{78}Ga 0.72	^{80}Ga 0.34	^{81}Ge -0.32	-
Kr	^{82}As 0.86	^{82}Ge 0.35	-	-
Rb	^{87}Se -0.51	^{85}Ge -0.45	-	-
Sr	^{88}Br -0.58	^{87}Br 0.35	-	-
Y	^{90}Br -0.54	^{89}Br -0.30	-	-
Zr	^{92}Kr -0.47	^{94}Rb -0.40	-	-
Nb	^{93}Kr -0.54	^{92}Kr 0.50	^{92}Rb 0.38	-
Mo	^{94}Rb 0.61	^{100}Y -0.45	-	-
Ru	^{104}Nb -0.57	-	-	-
Rh	^{103}Nb -0.45	^{103}Y -0.42	^{103}Zr -0.36	-
Pd	^{110}Tc -0.47	^{104}Nb 0.34	-	-
Ag	^{108}Tc 0.37	^{109}Tc -0.32	^{109}Mo -0.32	-
Cd	^{110}Tc 0.57	-	-	-
In	^{114}Rh 0.53	^{115}Rh -0.50	-	-
Sn	^{116}Rh 0.46	^{124}Ag -0.34	-	-
Sb	^{123}Ag -0.51	^{123}Cd -0.44	^{120}Ag 0.33	-
Te	^{130}In -0.62	^{129}Cd 0.33	-	-
I	^{127}Cd -0.71	-	-	-
Xe	^{130}In 0.62	^{129}Cd -0.38	^{129}Sn -0.32	-

Yttrium – ^{90}Br

^{90}Br has a 25% probability of beta-delayed neutron emission and indirectly impact **Yttrium** formation



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Reduction in uncertainties

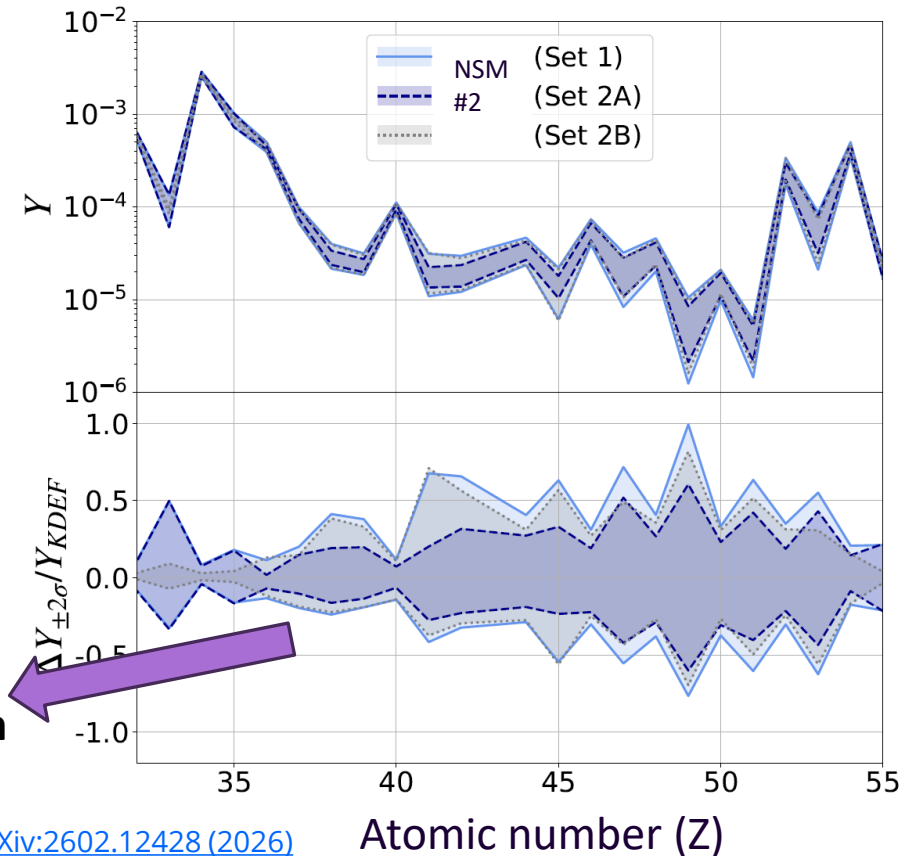
2σ (95 percentile) uncertainty bands
Hypothetical reduction in uncertainties

Set 1 : No reduction

Set 2A: Reduce uncertainties on rates for only **50** isotopes

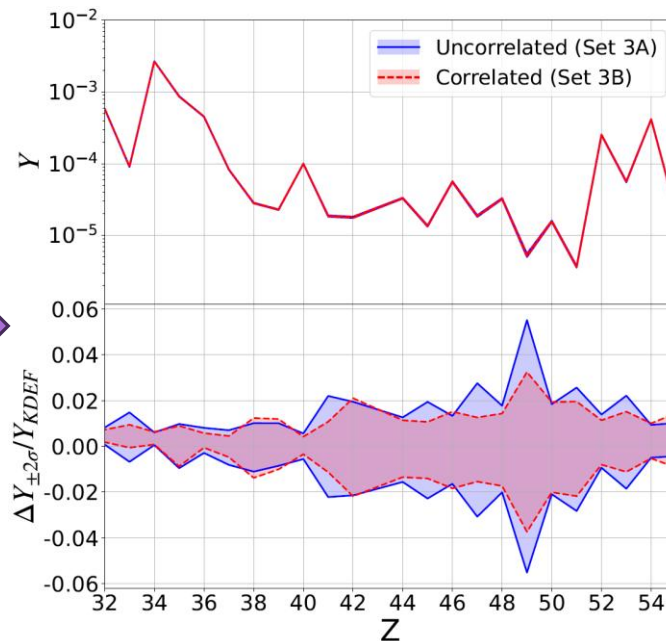
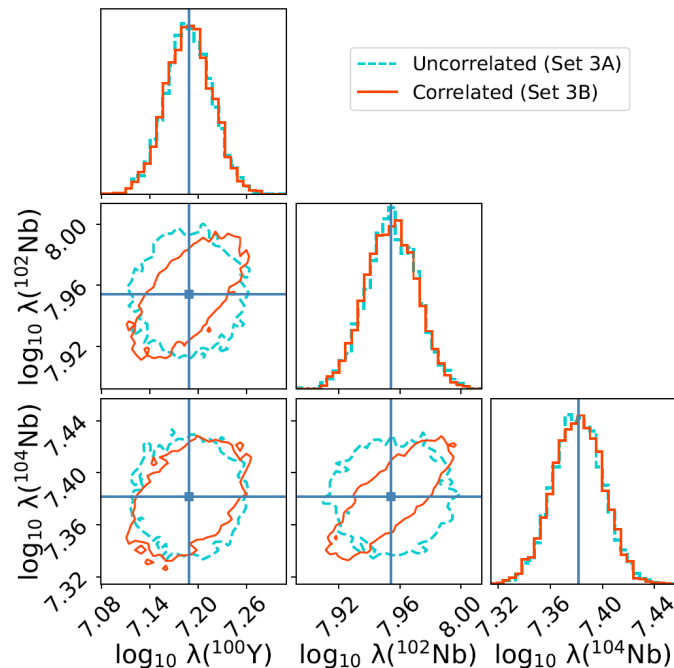
Set 2B: Reduce uncertainties on rates for all **1000** but not for the 50 isotopes.

The 50 isotopes lead to more reduction in uncertainty than the remaining 1000 do!!



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Theoretical constraints of Reaction rates



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

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1. Introduction

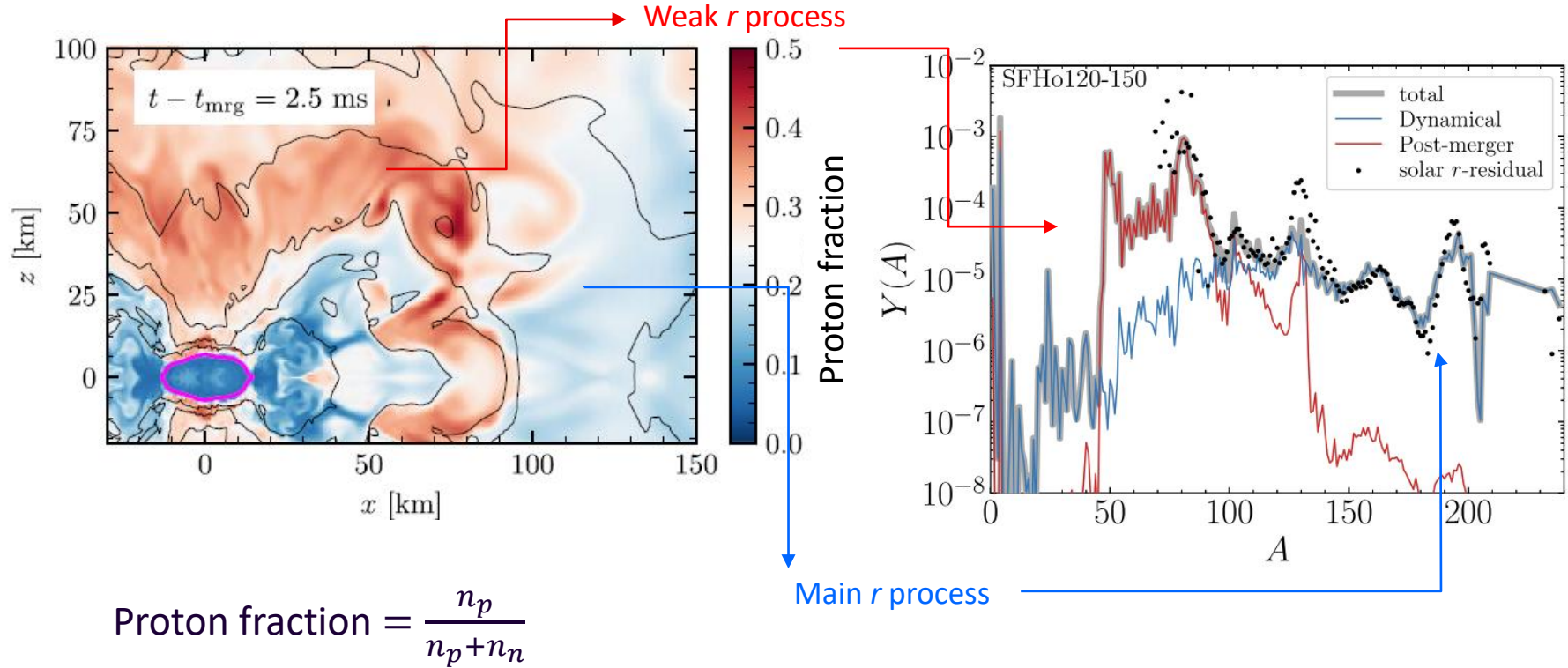
- Question: what is the astrophysical origin of *heavy* elements?
- Answer: Neutron star mergers?

2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)

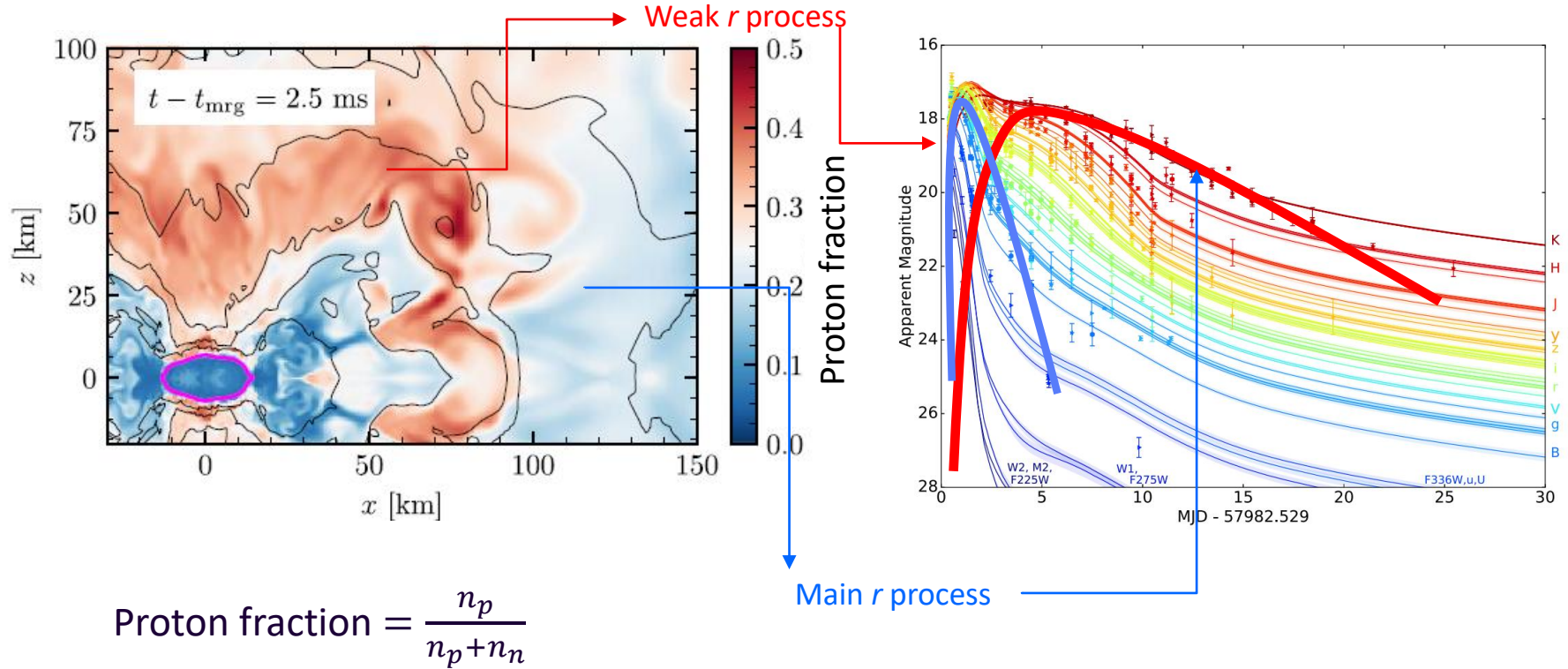
3. **Modelling Astrophysics: How much *r*-process is produced? (mass)**

4. Conclusions

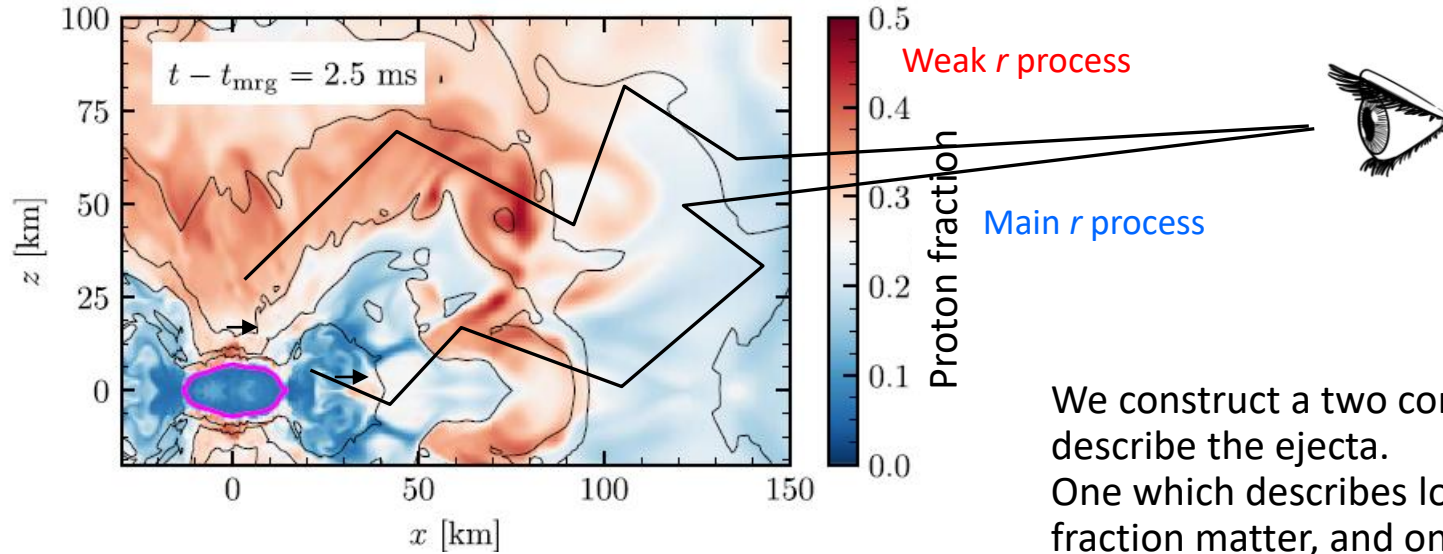
Abundance dependence on ejecta



Kilonova contribution of the ejecta



Kilonova contribution of the ejecta

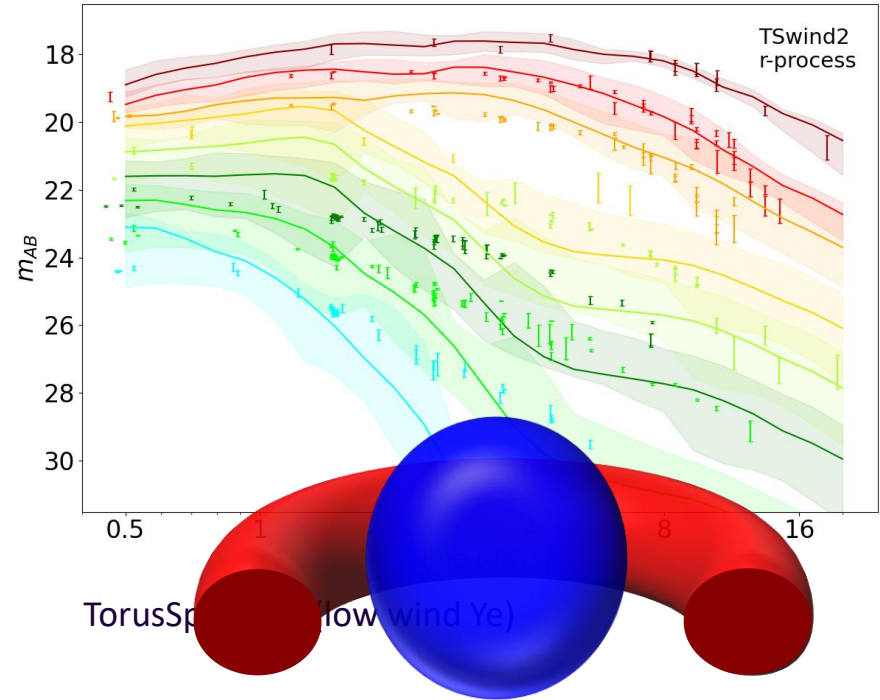
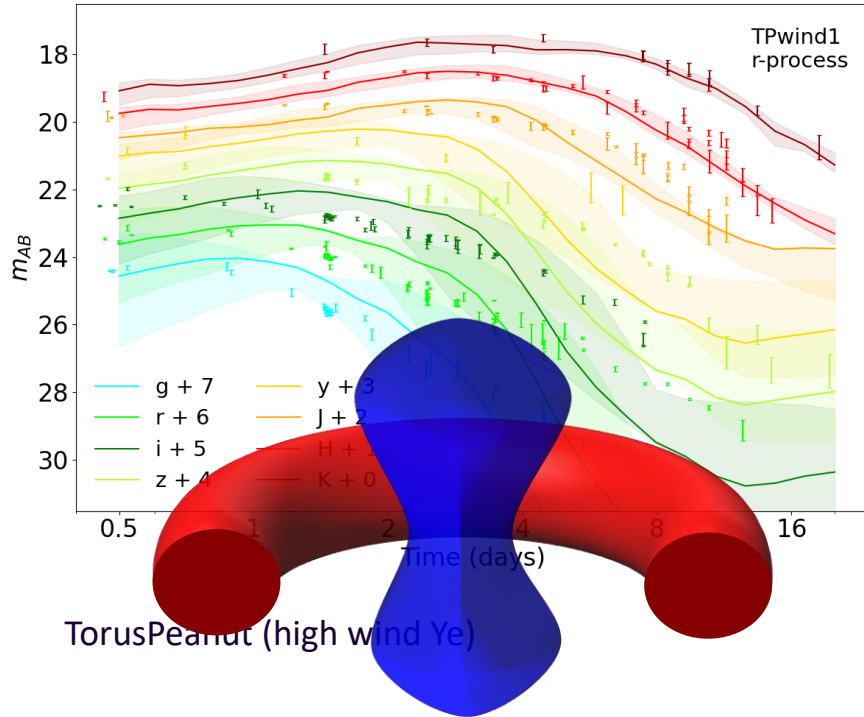


We construct a two component to describe the ejecta. One which describes low proton fraction matter, and one which has a high proton fraction matter.

Radiative transfer model

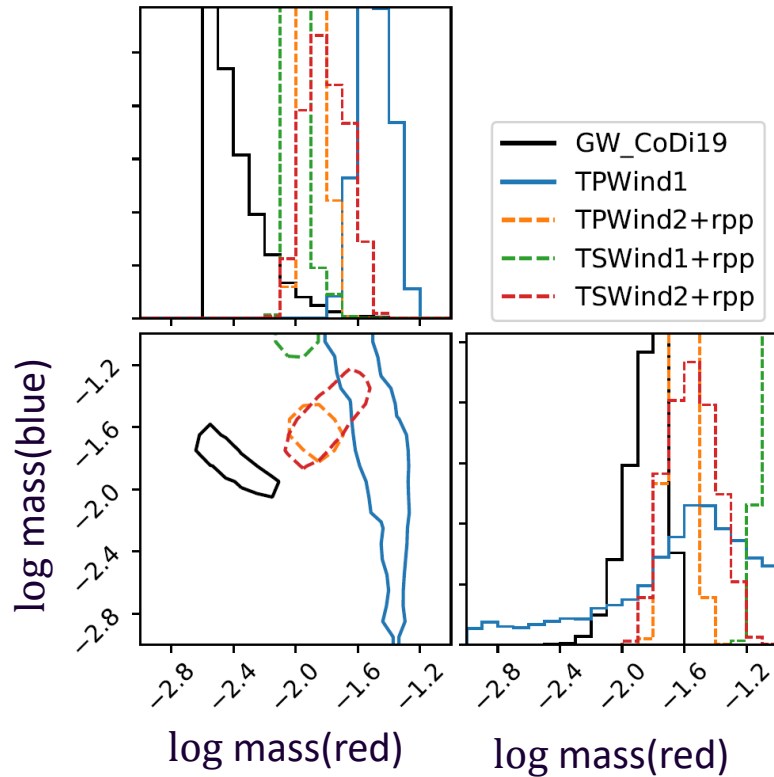
- Radiative transfer **SuperNu**. (Wollaeger et al 2013, 2014, Wollaeger, .., Kedia, Zenodo, 2023)
- Nucleosynthesis and heating from **WinNet**. (Winteler et al. 2012, Korobkin et al. 2012)
- Thermalization efficiencies (Barnes et al. (2016))
- Atomic opacities (Fontes et al. 2020)

Kilonova Light curves

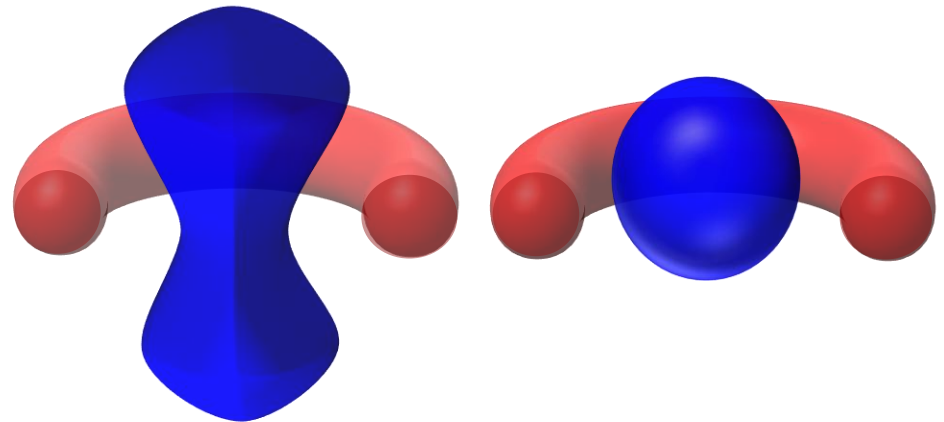


AK et al Phys. Rev. Research 5, 013168 (2023)

Bayesian inference based ejecta masses from GW170817

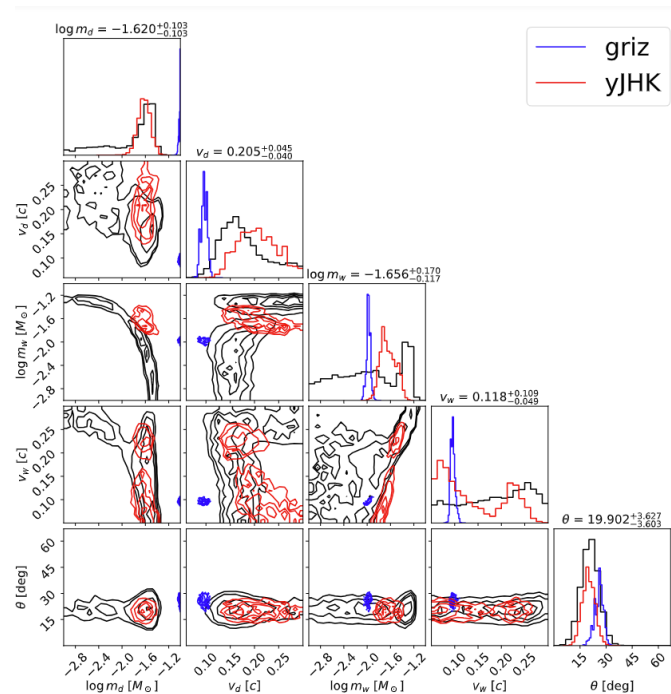
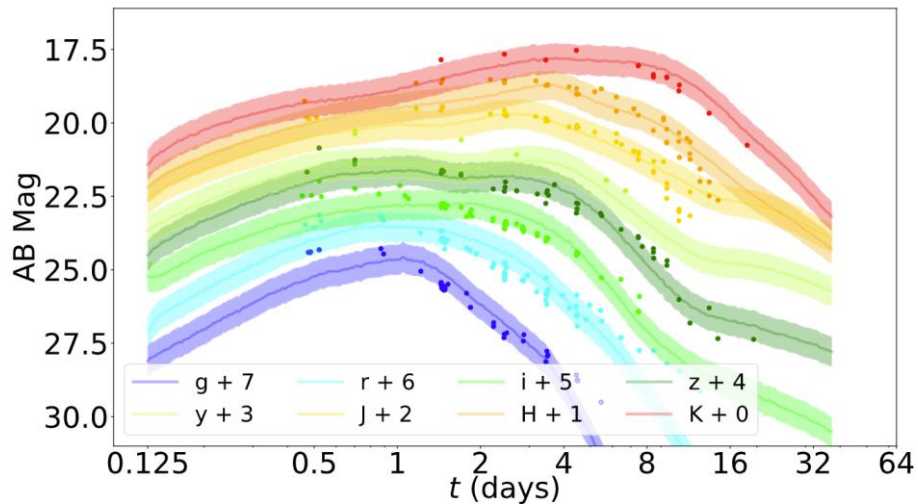


General agreement between the GW and kilonova observations on the inferred r -process enriched ejecta. However, how much mass was ejected depends on the morphology.



AK, et al, Phys. Rev. Res. 5, 013168 (2023);

Light curves w/ Neural Networks



Yinglei Peng*, Ristic, Kedia, et al. [Phys. Rev. Res. 6, 033078 \(2024\)](#)

*Conducted as undergrad at URochester, currently a graduate student at GeorgiaTech

Inference dependence on chosen wavebands.

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 - Question: what is the astrophysical origin of *heavy* elements?
 - **Answer: Neutron star mergers -> Maybe**
2. Modelling Nuclear physics: Which elements are synthesized in *r*-process? (proportionally)
3. Modelling Astrophysics: How much *r*-process is produced? (mass) (Answers: earlier slides)
4. Conclusions

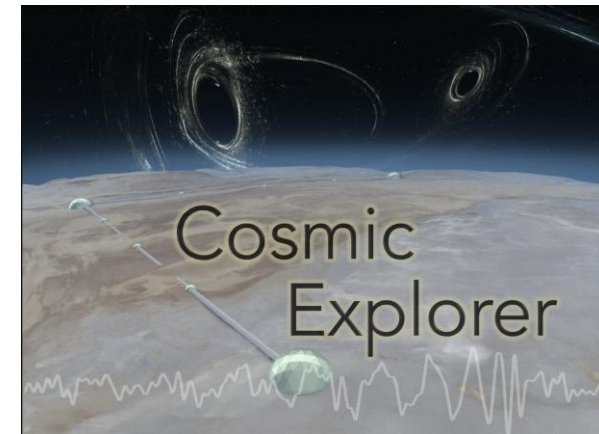
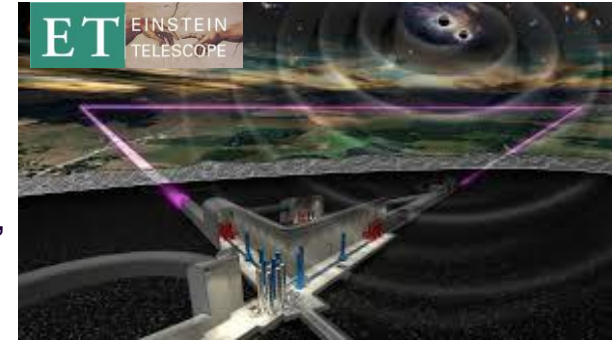
Future of observations and experiments



Nuclear experiments at:
TUNL (Triangle), FRIB (MSU),
ARIEL (TRIUMF)

EM observation:
Nancy Grace Roman Space Telescope,
Vera Rubin Observatory,
James Webb Space Telescope

GW observation:
LIGO-Virgo-KAGRA
3rd Generation GW detectors



Summary

- **Origin of elements**
- **Neutron star mergers — GW170817**
 - Supernovae
- ***r*-process nucleosynthesis**
 - Uncertainties in abundance from theoretical models
 - Isotopes that reduce uncertainties
- **Kilonova modelling**
 - Modelling in the two component (red, blue) method
 - Uncertainties in the amount of ejected mass produced.

Outlook

- ***r*-process nucleosynthesis**
 - Constraints on nuclear reaction rates
 - Astrophysical environments: Supernovae, Neutron star mergers, Collapsars, etc.
- **Kilonova modelling**
 - Apply hydrodynamic data into kilonova model
 - Model asymmetries such as due to neutron star-black hole merger.
 - Nuclear uncertainties in radioactivity

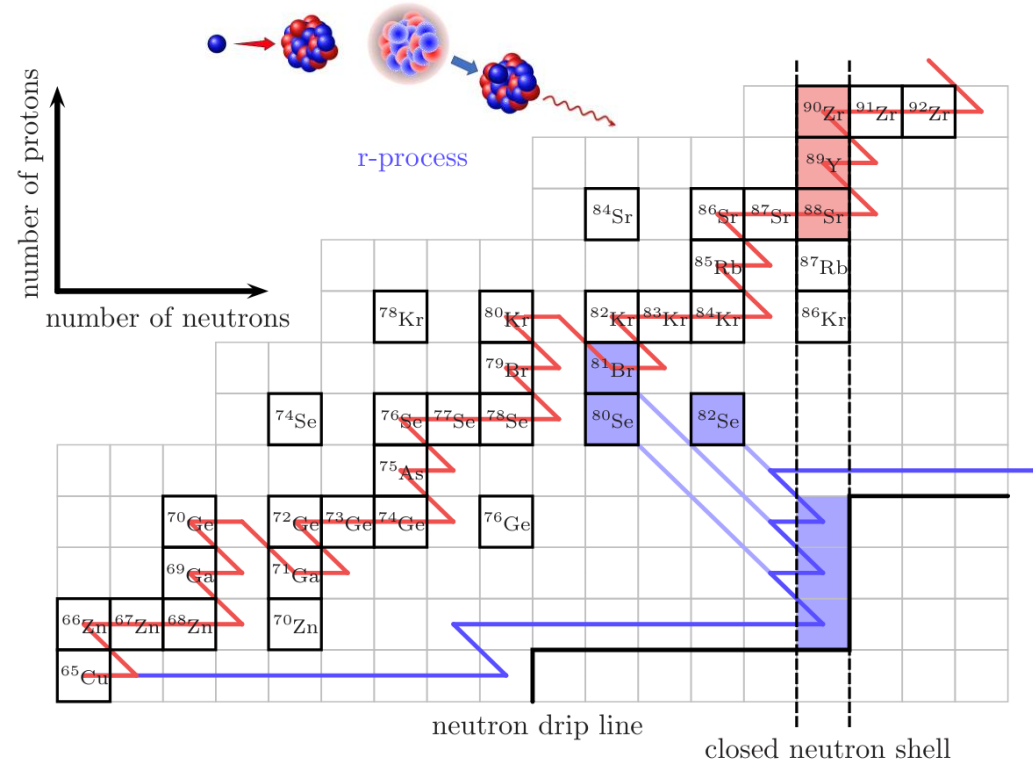
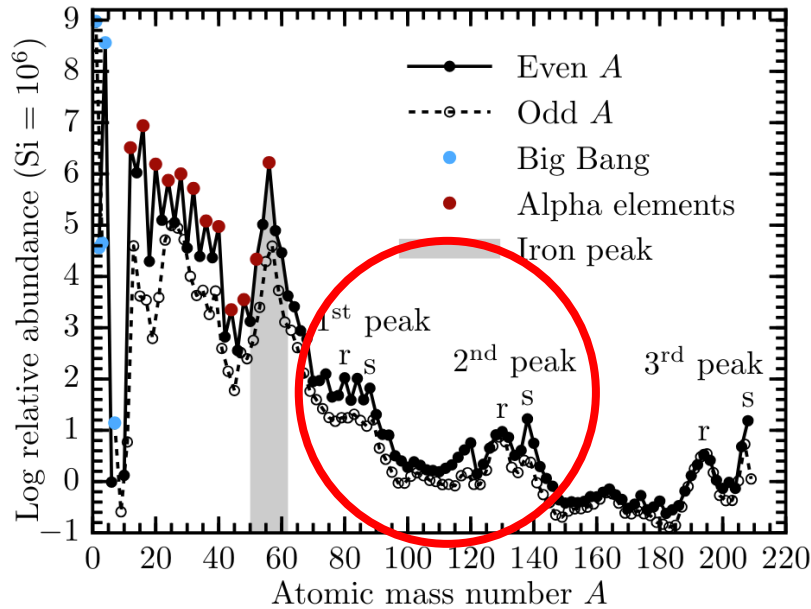
More theory, observational, and experimental work in nuclear astrophysics.



askedia@ncsu.edu

Extra slides

Solar chemical composition And the r process



Figures from [Lippuner \(2018\)](#)

r -process: $\tau_{n\text{-capture}} \ll \tau_{\beta\text{-decay}}$

Astrophysical sites for the r process

Binary Neutron star mergers:

(Meyer+1989, Metzger & Fernandez 14, Just+15, Radice+18, Kedia+22, Lund+24)

Neutron star-black hole mergers:

Surman+08, Darbha+21, Curtis+23

Magnetorotational supernovae:

Nishimura+15, Mosta+18, Reichert+21, Zha+24

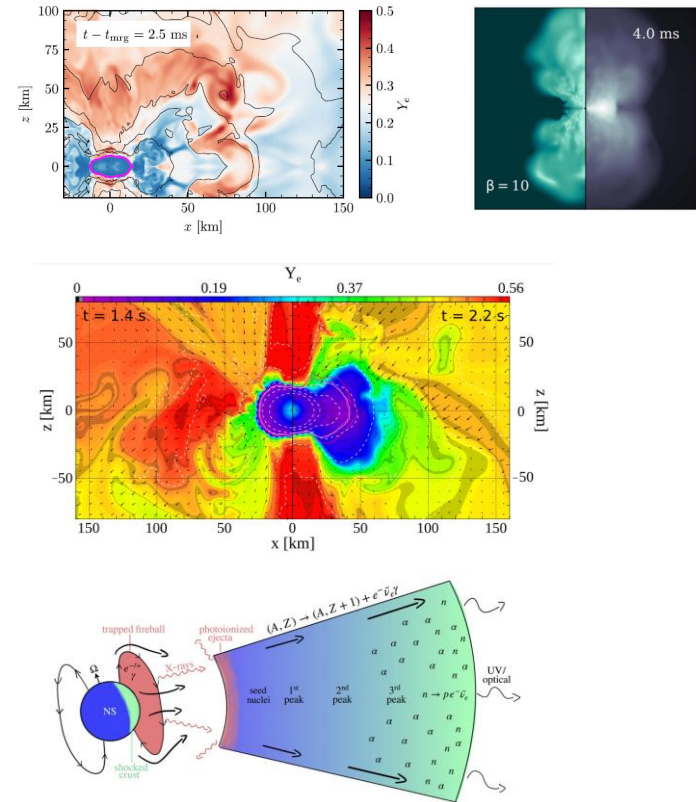
Magnetar giant flares:

Patel+25a, 25b

Collapsars:

Siegel+2019, Anand+2024

and **Others exotic scenarios**

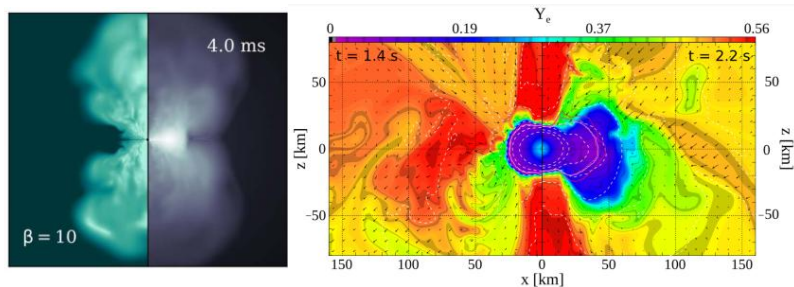


Astrophysics and nuclear physics inputs

Astrophysics:

Magnetized and non-magnetized disk winds from neutron star mergers
(K. Lund et al 2024, Metzger & Fernandez 2014)

Magnetorotational supernova
(Reichert et al 2021)

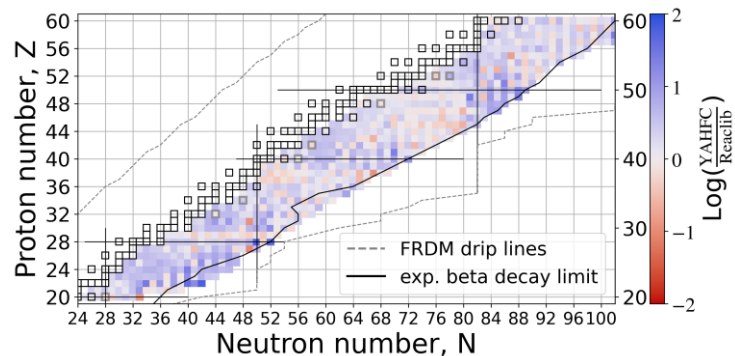


Nuclear physics:

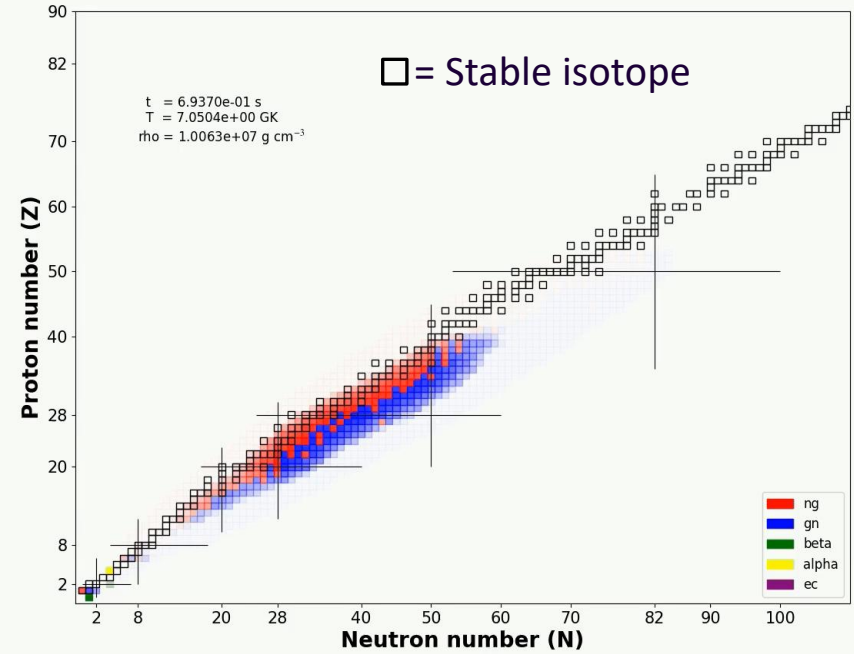
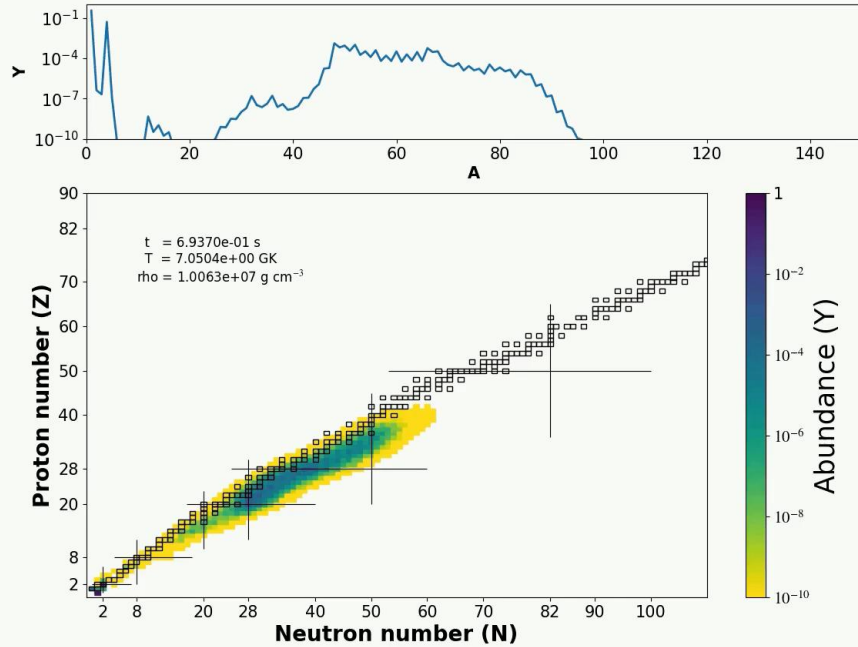
Neutron capture rates from YAHFC
(Yet Another Hauser-Feshbach Code)
(Ormand, Kravvaris, LLNL)

1000 isotopes, $Z=20-60$;

Beta-decay rates



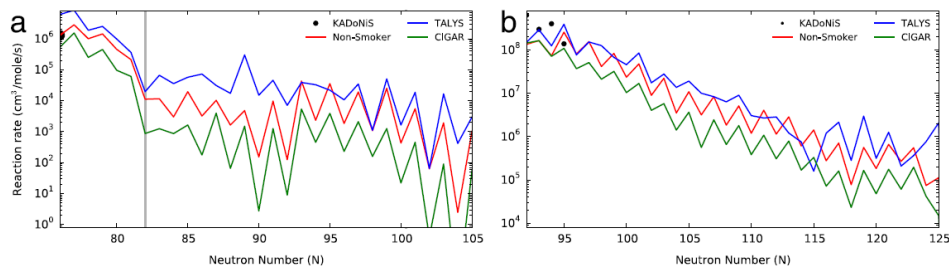
R-process evolution



Movies by AK; Calculation done on **PRISM** (reaction network code)

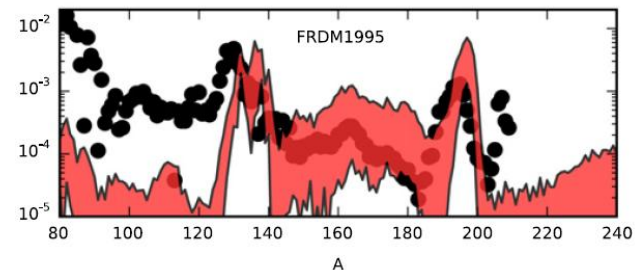
ng -> neutron capture
gn -> photodissociation (neutron removal)
beta -> beta decay

Uncertainties in neutron capture reactions



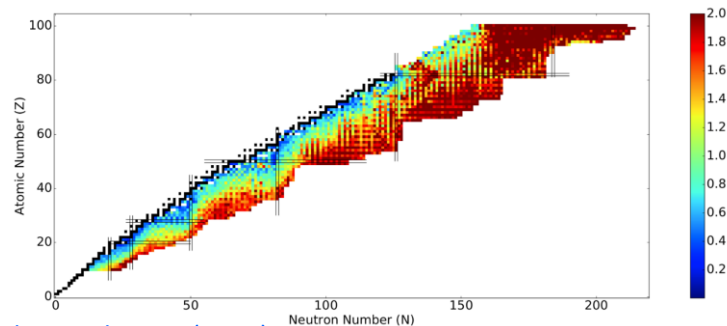
Sn –chain

Eu –chain



[Mumpower+, PNP \(2016\)](#) ;

multiplicative factors: $10^{-3} - 10^3$ in n-cap rates;
Mumpower et al ApJ 970:173 (2024)

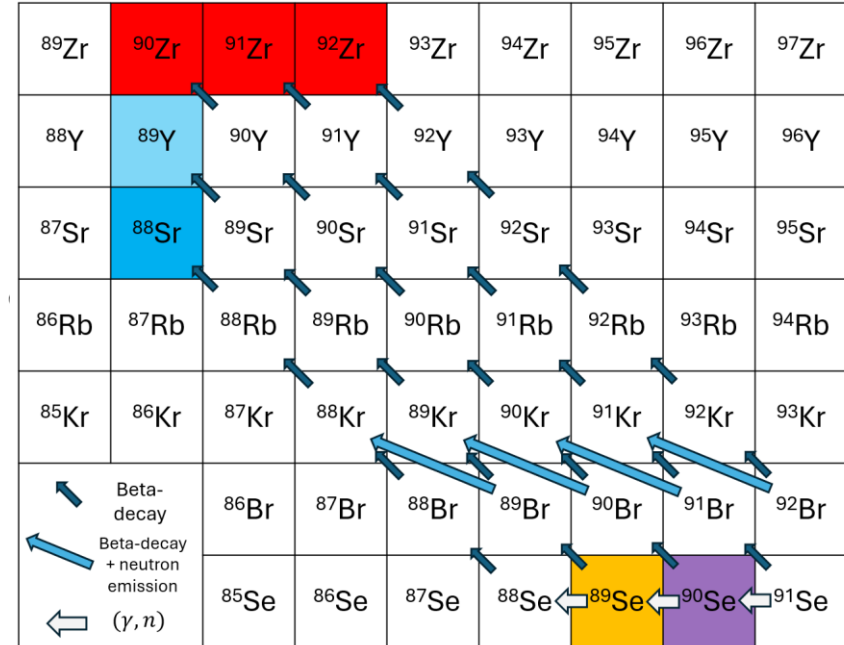
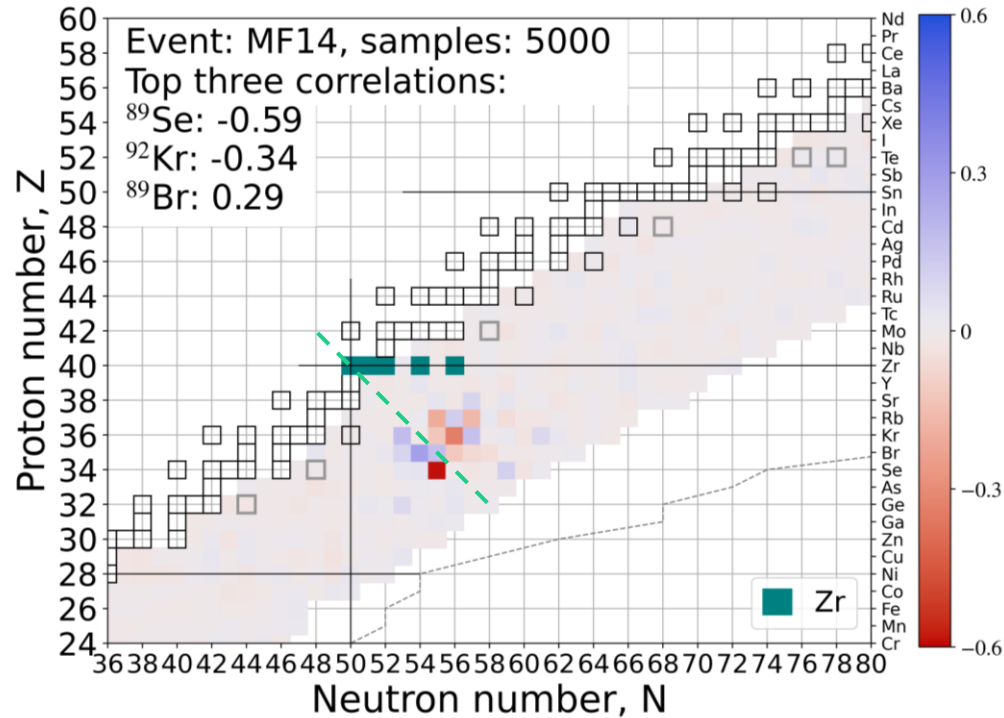


[Nikas et al, arXiv \(2020\)](#)

N-capture uncertainties

N-cap uncertainties propagated to abundances

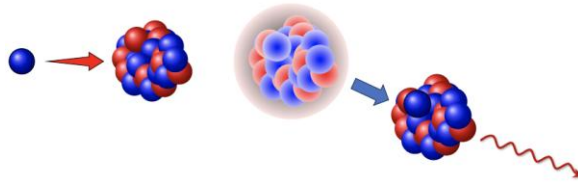
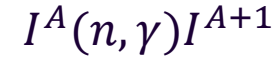
Elemental abundance — isotopic neutron capture rate correlation



Due to detailed balance: $\lambda_{\gamma, n}^{Z, A+1} \propto \lambda_{n, \gamma}^{Z, A}$

Nomenclature: Reactions at play in r -process

Neutron Capture



Photodissociation



Beta-decay



Hauser-Feshbach statistical model

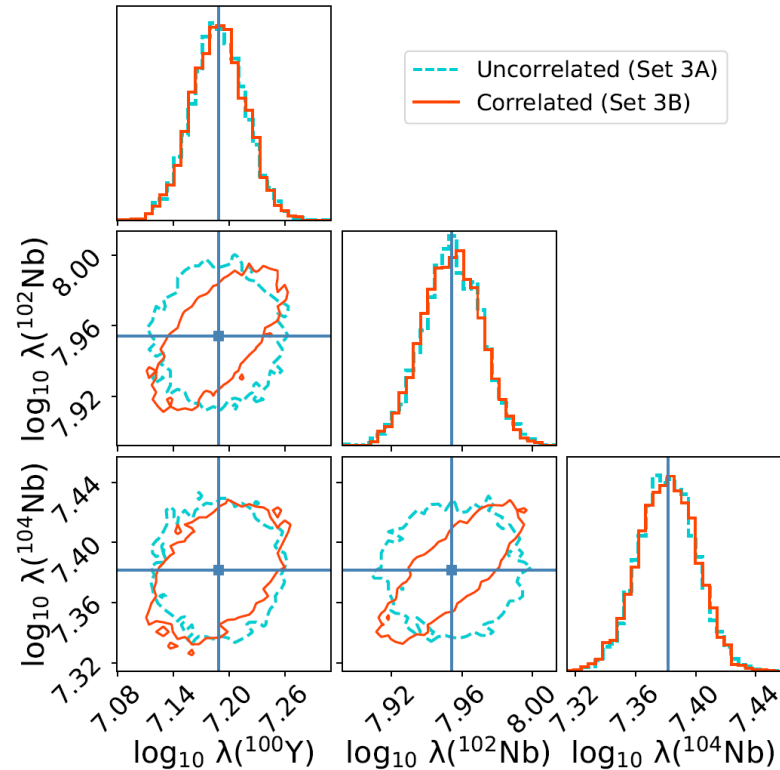
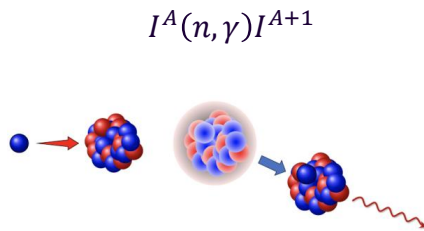
Hauser-Feshbach statistical model inputs:

☐ Mass Model

☒ Optical Model Potential

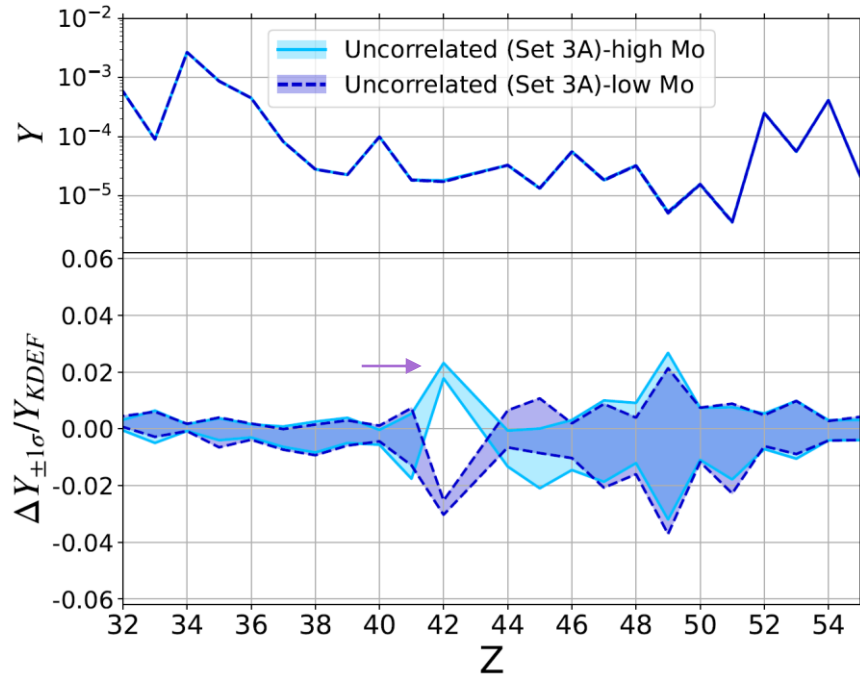
☐ Level Density

☐ Gamma-Strength Function

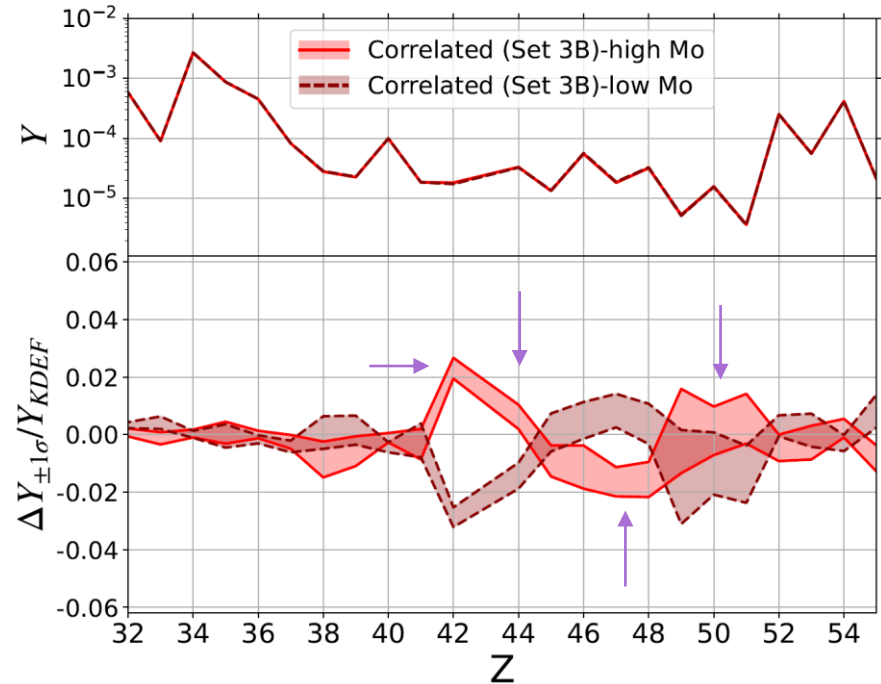


Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Correlated neutron capture rates



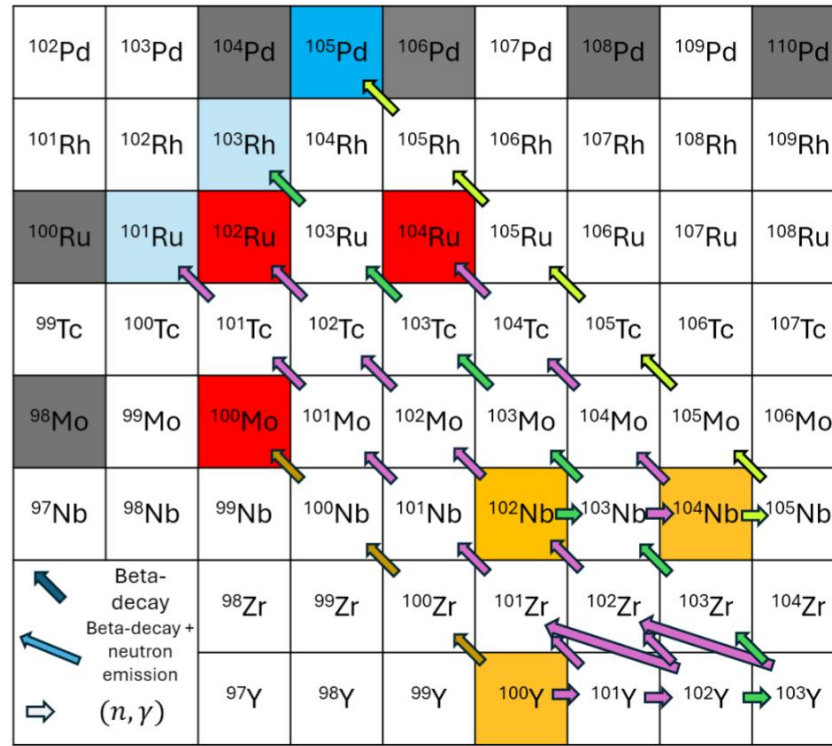
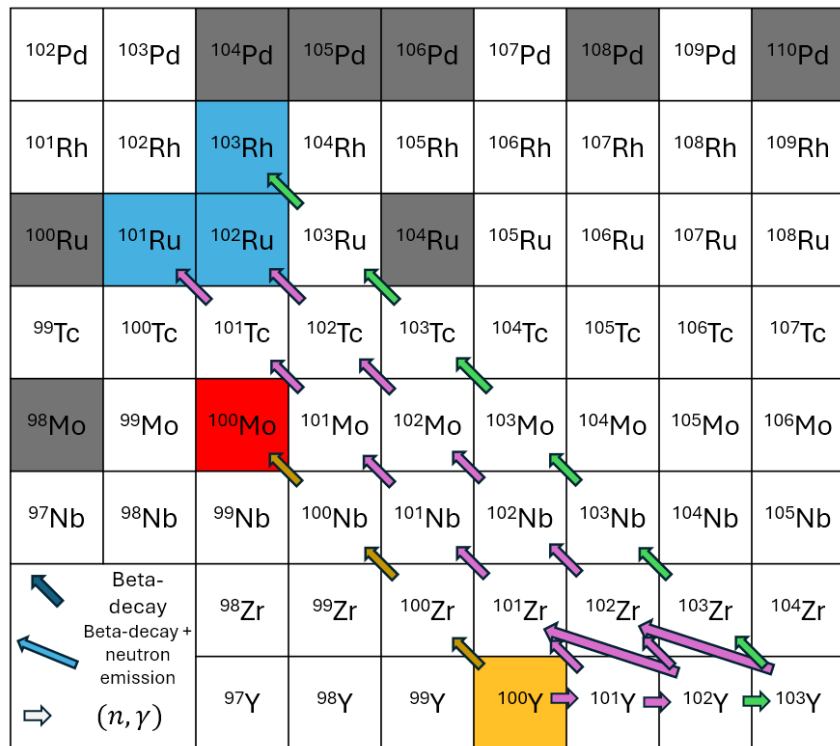
Uncorrelated: Simulations with high Mo (42) do not exhibit an increase or decrease in most other elements.



Correlated: Simulations with high Mo (42) also have high Ru (44), Sb(51) and low Rh (45), Pd (46), Ag (47), Cd (48).

Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Correlations in neutron capture rates – Toy model



Kedia, et al., [arXiv:2602.12428 \(2026\)](https://arxiv.org/abs/2602.12428)

Supernova 1987a

First Multi-Messenger event

Distance ~ 50 kpc

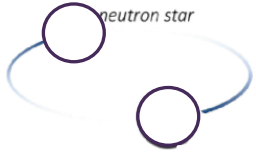
Electromagnetic (EM) and
neutrino emission detected



Bayesian inference of BNS ejecta

Binary properties

$$\vec{x} = \{M_{\text{BIL}}, M_{\text{NS}}, \chi_{\text{BIL}}, \Lambda_{\text{NS}}, \dots\}$$

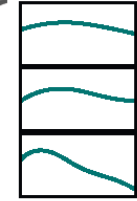
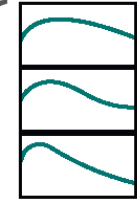


Outflow properties

$$\vec{y} = \{M_{\text{dyn}}, M_{\text{wind}}, v_{\text{dyn}}, v_{\text{wind}}, \dots\}$$



Light curves



Fit formulae to
numerical simulations

$$\vec{x}_1$$

BNS merger modelling
(Numerical relativity)

Radiative transfer
simulations

$$\vec{y}_{11}$$

$$\vec{y}_{12}$$

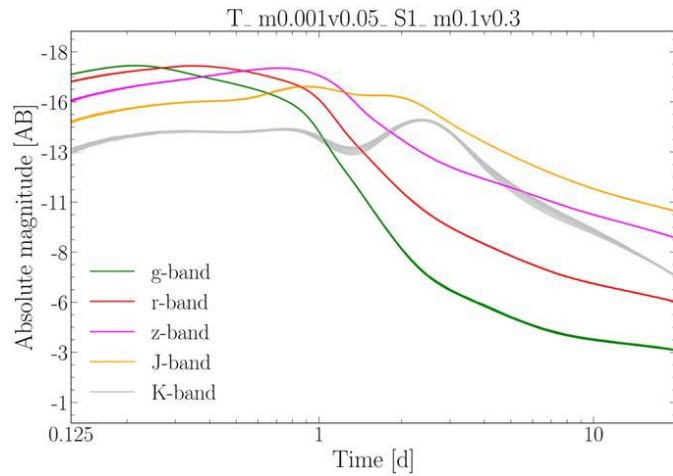
$$\vec{y}_{13}$$

$$\vdots$$

Ejecta material that
undergoes
nucleosynthesis

Kilonova
modelling

Adapted from Raaijmakers et al, [ApJ \(2021\)](#)



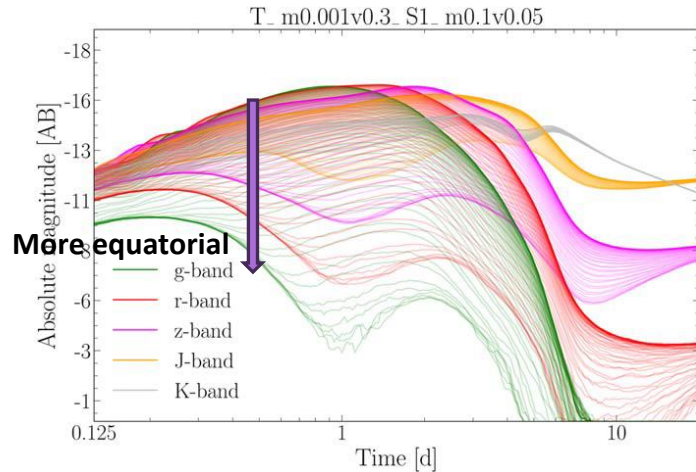
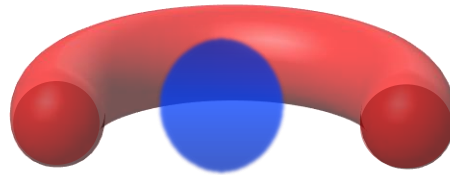
Lanthanide curtaining

A smoking gun for r-process

Curtaining of low wavelength (“blue”) light by Lanthanides (present in the “red” ejecta) produced by the nucleosynthesis.

Top panel: Ejecta more spherically symmetric and less obstructed by the “red” component. Therefore, there is virtually no angular dependence.

Bottom: “Red” component travels further and obstructs light emitted along the equator. Therefore, Lanthanide curtaining is observed. Low wavelength light is suppressed substantially along the equator. High wavelength light passes unaffected (spherically symmetric).

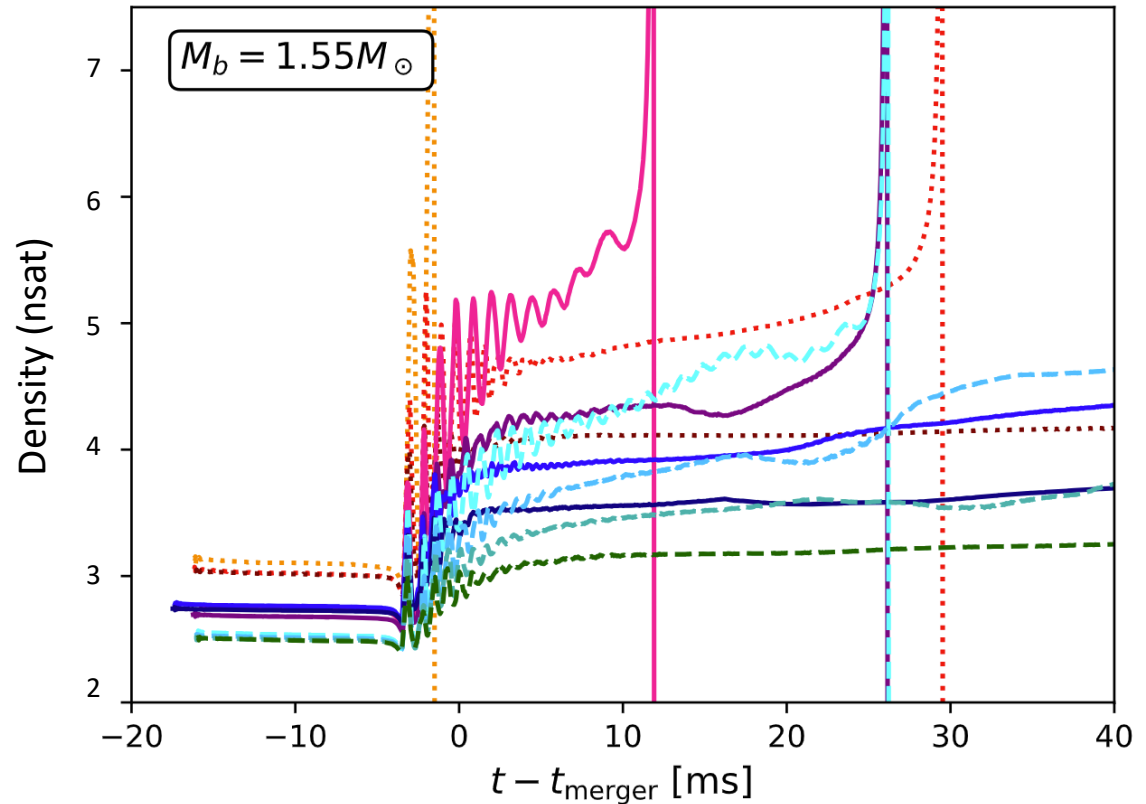


Wollaeger et al ApJ 2021; See also Korobkin+2021, Kasen+2015

Collapse of hypermassive NSs

Postmerger hypermassive
Neutron star remnant
dependence on the
nuclear Equation of State
(i.e. what defines the
neutron star structure)

Vertical drop represents
black hole formation.



Wei Sun, Mathews, Kedia, et al., (in prep)